Week 1:

Unix commands are a set of commands that are used to interact with the Unix operating system. Unix is a powerful, multi-user, multi-tasking operating system that was developed in the 1960s by Bell Labs. Unix commands are entered at the command prompt in a terminal window, and they allow users to perform a wide variety of tasks, such as managing files and directories, running processes, managing user accounts, and configuring network settings. Unix is now one of the most commonly used Operating systems used for various purposes such as Personal use, Servers, Smartphones, and many more. It was developed in the 1970's at AT& T Labs by two famous personalities Dennis M. Ritchie and Ken Thompson.

- the linux/unix commands are run in the terminal of a linux system.Terminal is like command prompt as that of in Windows OS)
- Linux/Unix commands are **case-sensitive** i.e Hello is different from hello**.**

Task-1

Hands on UNIX/Linux Commands

AIM: TO Execute Essential UNIX Commands

1. who : The '$ who' command displays all the users who have logged into the system currently.

```
$ who
```

```
Output:CSE tty2 2017-07-18 09:32 (:0)
```

2. pwd : The '$pwd' command stands for 'print working directory' and as the name says,it displays the directory in which we are currently.

```
$ pwd
```

```
Output: /home/CSE
```

3. mkdir : The '$ mkdir' stands for 'make directory' and it creates a new directory.

```
$ mkdir newfolder
```

4.cd : The '$ cd' command stands for 'change directory' and it changes your current directory to the 'newfolder' directory.

```
$ cd newfolder
```

5. ls : The 'ls' command simply displays the contents of a directory.

```
$ ls
```

```
Output: Desktop Documents Downloads Music Pictures Public Scratch Templates Videos
```

6. cp : This '$ cp ' command stands for 'copy' and it simply copy/paste the file wherever you want to.

```
$ cp /home/CSE/file.txt  /home/CSE/new.txt
```

7. mv : The '$ mv' command stands for 'move' and it simply move a file from a directory to another directory.

```
$ mv /home/CSE/file.txt  /home/CSE/new
```

\

8.rmdir : The '$ rmdir' command deletes any empty directory we want to delete.

```
$ rmdir newfolder
```

9. rm : The '$ rm ' command for remove file.

```
$ rm file.txt.
```

10. cal : The '$ cal' means calendar and it simply display calendar on to your screen.

```
$ cal
```

```
Output : July 2017

Su Mo Tu We Th Fr Sa

1

2 3 4 5 6 7 8

9 10 11 12 13 14 15

16 17 18 19 20 21 22

23 24 25 26 27 28 29

30 31
```

11. sort : As the name suggests the '$ sort' sorts the contents of the file according to the ASCII rules.

```
$ sort file
```

12. man : The '$ man' command stands for 'manual' and it can display the in-built manual for most of the commands that we ever need.In the above example, we can read about the '$ pwd' command.

```
$ man pwd
```

13. clear : The '$ clear' command is used to clean up the terminal.

```
$ clear
```

14.cat(concatenate) command is very frequently used in Linux. It reads data from the file and gives their content as output. It helps us to create, view, concatenate files.

1) To view a single file
Command:

```
$cat filename
```

Output

```
It will show content of given filename
```

2) To view multiple files
Command:

```
$cat file1 file2
```

Output

```
This will show the content of file1 and file2.
```

3) To view contents of a file preceding with line numbers.
Command:

```
$cat -n filename
```

4) Create a file
Command:

```
$ cat > newfile
```

5) Copy the contents of one file to another file.
Command:

```
$cat [filename-whose-contents-is-to-be-copied] > [destination-filename]
```

7) Cat command can append the contents of one file to the end of another file.
Command:

```
$cat file1 >> file2
```

grep

The grep filter searches a file for a particular pattern of characters, and displays all lines that contain that pattern. The pattern that is searched in the file is referred to as the regular expression (grep stands for global search for regular expression and print out).

Syntax:

```
grep [options] pattern [files]
```

```
Options Description
-c : This prints only a count of the lines that match a pattern
-h : Display the matched lines, but do not display the filenames.
-i : Ignores, case for matching
-l : Displays list of a filenames only.
-n : Display the matched lines and their line numbers.
-v : This prints out all the lines that do not matches the pattern
-e exp : Specifies expression with this option. Can use multiple times.
-f file : Takes patterns from file, one per line.
-E : Treats pattern as an extended regular expression (ERE)
-w : Match whole word
-o : Print only the matched parts of a matching line,
 with each such part on a separate output line.

-A n : Prints searched line and nlines after the result.
-B n : Prints searched line and n line before the result.
-C n : Prints searched line and n lines after before the result.
```

```
$cat > file.txt
```

```
unix is great os. unix was developed in Bell labs.

learn operating system.

Unix linux which one you choose.

uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```

1. Case insensitive search : The -i option enables to search for a string case insensitively in the given file. It matches the words like “UNIX”, “Unix”, “unix”.

```
$grep -i "UNix" file.txt
```

Output:

```
unix is great os. unix was developed in Bell labs.

Unix linux which one you choose.

uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```

2. Displaying the count of number of matches : We can find the number of lines that matches the given string/pattern

```
$grep -c "unix" file.txt
```

Output:

```
2
```

3. Display the file names that matches the pattern : We can just display the files that contains the given string/pattern.

```
$grep -l "unix" *
```

or

```
$grep -l "unix" f1.txt f2.txt f3.xt f4.txt
```

Output:

```
file.txt
```

4. Checking for the whole words in a file : By default, grep matches the given string/pattern even if it is found as a substring in a file. The -w option to grep makes it match only the whole words.

```
$ grep -w "unix" file.txt
```

Output:

```
unix is great os. unix was developed in Bell labs.

uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```

5. Displaying only the matched pattern : By default, grep displays the entire line which has the matched string. We can make the grep to display only the matched string by using the -o option.

```
$ grep -o "unix" file.txt
```

6. Show line number while displaying the output using grep -n : To show the line number of file with the line matched.

```
$ grep -n "unix" file.txt
```

Output:

```
1:unix is great os. unix is free os.

4:uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```